

41. An $n \times n$ matrix $\mathbf{A}$ is called **symmetric** if $\mathbf{A}^T = \mathbf{A}$; that is, if $a_{ij} = a_{ji}$, for all $i, j = 1, \dots, n$. Show that if $\mathbf{A}$ is an $n \times n$ matrix, then $\mathbf{A} + \mathbf{A}^T$ is a symmetric matrix.
42. Let $\mathbf{A}$ be an $m \times n$ matrix. Show that $\mathbf{A}^T \mathbf{A}$ is a symmetric $n \times n$ matrix and $\mathbf{A} \mathbf{A}^T$ is a symmetric $m \times m$ matrix (see Problem 41).
43. The **inner product** of two vectors is a generalization of the dot product, for vectors with complex entries. It is defined by

$$(\mathbf{x}, \mathbf{y}) := \sum_{i=1}^n x_i \bar{y}_i, \quad \text{where}$$

$\mathbf{x} = \text{col}(x_1, x_2, \dots, x_n)$, $\mathbf{y} = \text{col}(y_1, y_2, \dots, y_n)$ are complex vectors and the overbar denotes complex conjugation.

- (a) Show that $(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \bar{\mathbf{y}}$, where $\bar{\mathbf{y}} = \text{col}(\bar{y}_1, \bar{y}_2, \dots, \bar{y}_n)$.
- (b) Prove that for any $n \times 1$ vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ and any complex number λ , we have
- $$(\mathbf{x}, \mathbf{y}) = \overline{(\mathbf{y}, \mathbf{x})},$$
- $$(\mathbf{x}, \mathbf{y} + \mathbf{z}) = (\mathbf{x}, \mathbf{y}) + (\mathbf{x}, \mathbf{z}),$$
- $$(\lambda \mathbf{x}, \mathbf{y}) = \lambda (\mathbf{x}, \mathbf{y}), \quad (\mathbf{x}, \lambda \mathbf{y}) = \bar{\lambda} (\mathbf{x}, \mathbf{y}).$$

9.4 Linear Systems in Normal Form

In keeping with the introduction presented in Section 9.1, we say that a system of n linear differential equations is in **normal form** if it is expressed as

$$(1) \quad \mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t),$$

where $\mathbf{x}(t) = \text{col}(x_1(t), \dots, x_n(t))$, $\mathbf{f}(t) = \text{col}(f_1(t), \dots, f_n(t))$, and $\mathbf{A}(t) = [a_{ij}(t)]$ is an $n \times n$ matrix. As with a scalar linear differential equation, a system is called **homogeneous** when $\mathbf{f}(t) \equiv \mathbf{0}$; otherwise, it is called **nonhomogeneous**. When the elements of $\mathbf{A}$ are all constants, the system is said to have **constant coefficients**. Recall that an n th-order linear differential equation

$$(2) \quad y^{(n)}(t) + p_{n-1}(t)y^{(n-1)}(t) + \dots + p_0(t)y(t) = g(t)$$

can be rewritten as a first-order system in normal form using the substitution $x_1(t) := y(t)$, $x_2(t) := y'(t)$, $\dots$, $x_n(t) := y^{(n-1)}(t)$; indeed, equation (2) is equivalent to $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t)$, where $\mathbf{x}(t) = \text{col}(x_1(t), \dots, x_n(t))$, $\mathbf{f}(t) := \text{col}(0, \dots, 0, g(t))$, and

$$\mathbf{A}(t) := \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -p_0(t) & -p_1(t) & -p_2(t) & \cdots & -p_{n-2}(t) & -p_{n-1}(t) \end{bmatrix}.$$

The theory for systems in normal form parallels very closely the theory of linear differential equations presented in Chapters 4 and 6. In many cases the proofs for scalar linear differential equations carry over to normal systems with appropriate modifications. Conversely, results for normal systems apply to scalar linear equations since, as we showed, any scalar linear equation can be expressed as a normal system. This is the case with the existence and uniqueness theorems for linear differential equations.

The **initial value problem** for the normal system (1) is the problem of finding a differentiable vector function $\mathbf{x}(t)$ that satisfies the system on an interval I and also satisfies the **initial condition** $\mathbf{x}(t_0) = \mathbf{x}_0$, where t_0 is a given point of I and $\mathbf{x}_0 = \text{col}(x_{1,0}, \dots, x_{n,0})$ is a given vector.

Existence and Uniqueness

Theorem 2. If $\mathbf{A}(t)$ and $\mathbf{f}(t)$ are continuous on an open interval I that contains the point t_0 , then for any choice of the initial vector $\mathbf{x}_0$, there exists a unique solution $\mathbf{x}(t)$ on the whole interval I to the initial value problem

$$\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0.$$

We give a proof of this result in Chapter 13[†] and obtain as corollaries the existence and uniqueness theorems for second-order equations (Theorem 4, Section 4.5, page 182) and higher-order linear equations (Theorem 1, Section 6.1, page 319).

If we rewrite system (1) as $\mathbf{x}' - \mathbf{A}\mathbf{x} = \mathbf{f}$ and define the operator $L[\mathbf{x}] := \mathbf{x}' - \mathbf{A}\mathbf{x}$, then we can express system (1) in the operator form $L[\mathbf{x}] = \mathbf{f}$. Here the operator L maps vector functions into vector functions. Moreover, L is a *linear* operator in the sense that for any scalars a, b and differentiable vector functions $\mathbf{x}, \mathbf{y}$, we have

$$L[a\mathbf{x} + b\mathbf{y}] = aL[\mathbf{x}] + bL[\mathbf{y}].$$

The proof of this linearity follows from the properties of matrix multiplication (see Problem 27).

As a consequence of the linearity of L , if $\mathbf{x}_1, \dots, \mathbf{x}_n$ are solutions to the *homogeneous* system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, or $L[\mathbf{x}] = \mathbf{0}$ in operator notation, then any linear combination of these vectors, $c_1\mathbf{x}_1 + \dots + c_n\mathbf{x}_n$, is also a solution. Moreover, we will see that if the solutions $\mathbf{x}_1, \dots, \mathbf{x}_n$ are linearly independent, then *every* solution to $L[\mathbf{x}] = \mathbf{0}$ can be expressed as $c_1\mathbf{x}_1 + \dots + c_n\mathbf{x}_n$ for an appropriate choice of the constants $c_1, \dots, c_n$.

Linear Dependence of Vector Functions

Definition 1. The m vector functions $\mathbf{x}_1, \dots, \mathbf{x}_m$ are said to be **linearly dependent on an interval I** if there exist constants $c_1, \dots, c_m$, not all zero, such that

$$(3) \quad c_1\mathbf{x}_1(t) + \dots + c_m\mathbf{x}_m(t) = \mathbf{0}$$

for all t in I . If the vectors are not linearly dependent, they are said to be **linearly independent on I** .

Example 1 Show that the vector functions $\mathbf{x}_1(t) = \text{col}(e^t, 0, e^t)$, $\mathbf{x}_2(t) = \text{col}(3e^t, 0, 3e^t)$, and $\mathbf{x}_3(t) = \text{col}(t, 1, 0)$ are linearly dependent on $(-\infty, \infty)$.

Solution Notice that $\mathbf{x}_2$ is just 3 times $\mathbf{x}_1$ and therefore $3\mathbf{x}_1(t) - \mathbf{x}_2(t) + 0 \cdot \mathbf{x}_3(t) = \mathbf{0}$ for all t . Hence, $\mathbf{x}_1, \mathbf{x}_2$, and $\mathbf{x}_3$ are linearly dependent on $(-\infty, \infty)$. ♦

Example 2 Show that

$$\mathbf{x}_1(t) = \begin{bmatrix} t \\ |t| \end{bmatrix}, \quad \mathbf{x}_2(t) = \begin{bmatrix} |t| \\ t \end{bmatrix}$$

are linearly independent on $(-\infty, \infty)$.

[†]All references to Chapters 11–13 refer to the expanded text, *Fundamentals of Differential Equations and Boundary Value Problems*, 7th ed.

Solution Note that at every instant t_0 , the column vector $\mathbf{x}_1(t_0)$ is a multiple of $\mathbf{x}_2(t_0)$; indeed, $\mathbf{x}_1(t_0) = \mathbf{x}_2(t_0)$ for $t_0 \geq 0$, and $\mathbf{x}_1(t_0) = -\mathbf{x}_2(t_0)$ for $t_0 \leq 0$. Nonetheless, the vector functions are not dependent, because the c 's in condition (3) are not allowed to change with t ; for $t < 0$, the equation $c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) = \mathbf{0}$ implies $c_1 - c_2 = 0$, but for $t > 0$ it implies $c_1 + c_2 = 0$. Thus $c_1 = c_2 = 0$ and the functions are independent. ♦

Example 3 Show that the vector functions $\mathbf{x}_1(t) = \text{col}(e^{2t}, 0, e^{2t})$, $\mathbf{x}_2(t) = \text{col}(e^{2t}, e^{2t}, -e^{2t})$, and $\mathbf{x}_3(t) = \text{col}(e^t, 2e^t, e^t)$ are linearly independent on $(-\infty, \infty)$.

Solution To prove independence, we *assume* c_1 , c_2 , and c_3 are constants for which

$$c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + c_3\mathbf{x}_3(t) = \mathbf{0}$$

holds at every t in $(-\infty, \infty)$ and show that this forces $c_1 = c_2 = c_3 = 0$. In particular, when $t = 0$ we obtain

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \mathbf{0},$$

which is equivalent to the system of linear equations

$$\begin{aligned} c_1 + c_2 + c_3 &= 0, \\ (4) \quad c_2 + 2c_3 &= 0, \\ c_1 - c_2 + c_3 &= 0. \end{aligned}$$

Either by solving (4) or by checking that the determinant of its coefficients is nonzero (recall Theorem 1 on page 511), we can verify that (4) has only the trivial solution $c_1 = c_2 = c_3 = 0$. Therefore the vector functions $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ are linearly independent on $(-\infty, \infty)$ (in fact, on any interval containing $t = 0$). ♦

As Example 3 illustrates, if $\mathbf{x}_1(t), \mathbf{x}_2(t), \dots, \mathbf{x}_n(t)$ are n vector functions, each having n components, we can establish their linear independence on an interval I if we can find *one* point t_0 in I where the determinant

$$\det[\mathbf{x}_1(t_0) \dots \mathbf{x}_n(t_0)]$$

is not zero. Because of the analogy with scalar equations, we call this determinant the **Wronskian**.

Wronskian

Definition 2. The **Wronskian** of n vector functions $\mathbf{x}_1(t) = \text{col}(x_{1,1}, \dots, x_{1,n})$, $\dots$, $\mathbf{x}_n(t) = \text{col}(x_{n,1}, \dots, x_{n,n})$ is defined to be the function

$$W[\mathbf{x}_1, \dots, \mathbf{x}_n](t) := \begin{vmatrix} x_{1,1}(t) & x_{1,2}(t) & \cdots & x_{1,n}(t) \\ x_{2,1}(t) & x_{2,2}(t) & \cdots & x_{2,n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1}(t) & x_{n,2}(t) & \cdots & x_{n,n}(t) \end{vmatrix}.$$

We see that n vector functions are linearly independent on an interval if their Wronskian is nonzero *at any point in the interval*. But now we show that if these functions happen to be independent solutions to a homogeneous system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is an $n \times n$ matrix of continuous functions, then the Wronskian is *never* zero on I . For suppose to the contrary that $W(t_0) = 0$. Then by Theorem 1 the vanishing of the determinant implies that the column vectors $\mathbf{x}_1(t_0), \mathbf{x}_2(t_0), \dots, \mathbf{x}_n(t_0)$ are linearly dependent. Thus there exist scalars $c_1, \dots, c_n$ *not all zero*, such that at t_0

$$c_1\mathbf{x}_1(t_0) + \cdots + c_n\mathbf{x}_n(t_0) = \mathbf{0}.$$

However, $c_1\mathbf{x}_1(t) + \cdots + c_n\mathbf{x}_n(t)$ and the vector function $\mathbf{z}(t) \equiv \mathbf{0}$ are both solutions to $\mathbf{x}' = \mathbf{A}\mathbf{x}$ on I , and they agree at the point t_0 . So these solutions must be identical on I according to the existence-uniqueness theorem (Theorem 2, page 516). That is,

$$c_1\mathbf{x}_1(t) + \cdots + c_n\mathbf{x}_n(t) = \mathbf{0}$$

for all t in I . But this contradicts the given information that $\mathbf{x}_1, \dots, \mathbf{x}_n$ are linearly independent on I . We have shown that $W(t_0) \neq 0$, and since t_0 is an arbitrary point, it follows that $W(t) \neq 0$ for all $t \in I$.

The preceding argument has two important implications that parallel the scalar case. First, the Wronskian of solutions to $\mathbf{x}' = \mathbf{A}\mathbf{x}$ is *either identically zero or never zero on I* (see also Problem 33). Second, a set of n solutions $\mathbf{x}_1, \dots, \mathbf{x}_n$ to $\mathbf{x}' = \mathbf{A}\mathbf{x}$ on I is *linearly independent on I if and only if their Wronskian is never zero on I* . With these facts in hand, we can imitate the proof given for the scalar case in Section 6.1 (Theorem 2, page 322) to obtain the following representation theorem for the solutions to $\mathbf{x}' = \mathbf{A}\mathbf{x}$.

Representation of Solutions (Homogeneous Case)

Theorem 3. Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be n linearly independent solutions to the homogeneous system

$$(5) \quad \mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$$

on the interval I , where $\mathbf{A}(t)$ is an $n \times n$ matrix function continuous on I . Then every solution to (5) on I can be expressed in the form

$$(6) \quad \mathbf{x}(t) = c_1\mathbf{x}_1(t) + \cdots + c_n\mathbf{x}_n(t),$$

where $c_1, \dots, c_n$ are constants.

A set of solutions $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ that are linearly independent on I or, equivalently, whose Wronskian does not vanish on I , is called a **fundamental solution set** for (5) on I . The linear combination in (6), written with arbitrary constants, is referred to as a **general solution** to (5).

If we take the vectors in a fundamental solution set and let them form the columns of a matrix $\mathbf{X}(t)$, that is,

$$\mathbf{X}(t) = [\mathbf{x}_1(t) \ \mathbf{x}_2(t) \ \cdots \ \mathbf{x}_n(t)] = \begin{bmatrix} x_{1,1}(t) & x_{1,2}(t) & \cdots & x_{1,n}(t) \\ x_{2,1}(t) & x_{2,2}(t) & \cdots & x_{2,n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1}(t) & x_{n,2}(t) & \cdots & x_{n,n}(t) \end{bmatrix},$$

then the matrix $\mathbf{X}(t)$ is called a **fundamental matrix** for (5). We can use it to express the general solution (6) as

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{c},$$

where $\mathbf{c} = \text{col}(c_1, \dots, c_n)$ is an arbitrary constant vector. Since $\det \mathbf{X} = W[\mathbf{x}_1, \dots, \mathbf{x}_n]$ is never zero on I , it follows from Theorem 1 on page 511 that $\mathbf{X}(t)$ is invertible for every t in I .

Example 4 Verify that the set

$$S = \left\{ \begin{bmatrix} e^{2t} \\ e^{2t} \\ e^{2t} \end{bmatrix}, \begin{bmatrix} -e^{-t} \\ 0 \\ e^{-t} \end{bmatrix}, \begin{bmatrix} 0 \\ e^{-t} \\ -e^{-t} \end{bmatrix} \right\}$$

is a fundamental solution set for the system

$$(7) \quad \mathbf{x}'(t) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \mathbf{x}(t)$$

on the interval $(-\infty, \infty)$ and find a fundamental matrix for (7). Also determine a general solution for (7).

Solution Substituting the first vector in the set S into the right-hand side of (7) gives

$$\mathbf{Ax} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} e^{2t} \\ e^{2t} \\ e^{2t} \end{bmatrix} = \begin{bmatrix} 2e^{2t} \\ 2e^{2t} \\ 2e^{2t} \end{bmatrix} = \mathbf{x}'(t).$$

Hence this vector satisfies system (7) for all t . Similar computations verify that the remaining vectors in S are also solutions to (7) on $(-\infty, \infty)$. For us to show that S is a fundamental solution set, it is enough to observe that the Wronskian

$$W(t) = \begin{vmatrix} e^{2t} & -e^{-t} & 0 \\ e^{2t} & 0 & e^{-t} \\ e^{2t} & e^{-t} & -e^{-t} \end{vmatrix} = e^{2t} \begin{vmatrix} 0 & e^{-t} \\ e^{-t} & -e^{-t} \end{vmatrix} + e^{-t} \begin{vmatrix} e^{2t} & e^{-t} \\ e^{2t} & -e^{-t} \end{vmatrix} = -3$$

is never zero.

A fundamental matrix $\mathbf{X}(t)$ for (7) is just the matrix we used to compute the Wronskian; that is,

$$(8) \quad \mathbf{X}(t) := \begin{bmatrix} e^{2t} & -e^{-t} & 0 \\ e^{2t} & 0 & e^{-t} \\ e^{2t} & e^{-t} & -e^{-t} \end{bmatrix}.$$

A general solution to (7) can now be expressed as

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{c} = c_1 \begin{bmatrix} e^{2t} \\ e^{2t} \\ e^{2t} \end{bmatrix} + c_2 \begin{bmatrix} -e^{-t} \\ 0 \\ e^{-t} \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ e^{-t} \\ -e^{-t} \end{bmatrix}. \quad \blacklozenge$$

It is easy to check that the fundamental matrix in (8) satisfies the equation

$$\mathbf{X}'(t) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \mathbf{X}(t);$$

indeed, this is equivalent to showing that $\mathbf{x}' = \mathbf{Ax}$ for each column $\mathbf{x}$ in S . In general, a fundamental matrix for a system $\mathbf{x}' = \mathbf{Ax}$ satisfies the corresponding **matrix differential equation** $\mathbf{X}' = \mathbf{AX}$.

Another consequence of the linearity of the operator L defined by $L[\mathbf{x}] := \mathbf{x}' - \mathbf{Ax}$ is the **superposition principle** for linear systems. It states that if $\mathbf{x}_1$ and $\mathbf{x}_2$ are solutions, respectively, to the nonhomogeneous systems

$$L[\mathbf{x}] = \mathbf{g}_1 \text{ and } L[\mathbf{x}] = \mathbf{g}_2,$$

then $c_1\mathbf{x}_1 + c_2\mathbf{x}_2$ is a solution to

$$L[\mathbf{x}] = c_1\mathbf{g}_1 + c_2\mathbf{g}_2.$$

Using the superposition principle and the representation theorem for homogeneous systems, we can prove the following theorem.

Representation of Solutions (Nonhomogeneous Case)

Theorem 4. If $\mathbf{x}_p$ is a particular solution to the nonhomogeneous system

$$(9) \quad \mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t)$$

on the interval I and $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is a fundamental solution set on I for the corresponding homogeneous system $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$, then every solution to (9) on I can be expressed in the form

$$(10) \quad \mathbf{x}(t) = \mathbf{x}_p(t) + c_1\mathbf{x}_1(t) + \dots + c_n\mathbf{x}_n(t),$$

where $c_1, \dots, c_n$ are constants.

The proof of this theorem is almost identical to the proofs of Theorem 4 in Section 4.5 (page 182) and Theorem 4 in Section 6.1 (page 325). We leave the proof as an exercise.

The linear combination of $\mathbf{x}_p, \mathbf{x}_1, \dots, \mathbf{x}_n$ in (10) written with arbitrary constants $c_1, \dots, c_n$ is called a **general solution** of (9). This general solution can also be expressed as $\mathbf{x} = \mathbf{x}_p + \mathbf{X}\mathbf{c}$, where $\mathbf{X}$ is a fundamental matrix for the homogeneous system and $\mathbf{c}$ is an arbitrary constant vector.

We now summarize the results of this section as they apply to the problem of finding a general solution to a system of n linear first-order differential equations in normal form.

Approach to Solving Normal Systems

1. To determine a general solution to the $n \times n$ homogeneous system $\mathbf{x}' = \mathbf{A}\mathbf{x}$:

- (a) Find a fundamental solution set $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ that consists of n linearly independent solutions to the homogeneous system.
- (b) Form the linear combination

$$\mathbf{x} = \mathbf{X}\mathbf{c} = c_1\mathbf{x}_1 + \dots + c_n\mathbf{x}_n,$$

where $\mathbf{c} = \text{col}(c_1, \dots, c_n)$ is any constant vector and $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]$ is the fundamental matrix, to obtain a general solution.

2. To determine a general solution to the nonhomogeneous system $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f}$:

- (a) Find a particular solution $\mathbf{x}_p$ to the nonhomogeneous system.
- (b) Form the sum of the particular solution and the general solution $\mathbf{X}\mathbf{c} = c_1\mathbf{x}_1 + \dots + c_n\mathbf{x}_n$ to the corresponding homogeneous system in part 1,

$$\mathbf{x} = \mathbf{x}_p + \mathbf{X}\mathbf{c} = \mathbf{x}_p + c_1\mathbf{x}_1 + \dots + c_n\mathbf{x}_n,$$

to obtain a general solution to the given system.

We devote the rest of this chapter to methods for finding fundamental solution sets for homogeneous systems and particular solutions for nonhomogeneous systems.

9.4 EXERCISES

In Problems 1–4, write the given system in the matrix form $\mathbf{x}' = \mathbf{Ax} + \mathbf{f}$.

1. $x'(t) = 3x(t) - y(t) + t^2$,
 $y'(t) = -x(t) + 2y(t) + e^t$
2. $r'(t) = 2r(t) + \sin t$,
 $\theta'(t) = r(t) - \theta(t) + 1$
3. $\frac{dx}{dt} = t^2x - y - z + t$,
 $\frac{dy}{dt} = e^tz + 5$,
 $\frac{dz}{dt} = tx - y + 3z - e^t$
4. $\frac{dx}{dt} = x + y + z$,
 $\frac{dy}{dt} = 2x - y + 3z$,
 $\frac{dz}{dt} = x + 5z$

In Problems 5–8, rewrite the given scalar equation as a first-order system in normal form. Express the system in the matrix form $\mathbf{x}' = \mathbf{Ax} + \mathbf{f}$.

5. $y''(t) - 3y'(t) - 10y(t) = \sin t$
6. $x''(t) + x(t) = t^2$
7. $\frac{d^4w}{dt^4} + w = t^2$
8. $\frac{d^3y}{dt^3} - \frac{dy}{dt} + y = \cos t$

In Problems 9–12, write the given system as a set of scalar equations.

9. $\mathbf{x}' = \begin{bmatrix} 5 & 0 \\ -2 & 4 \end{bmatrix} \mathbf{x} + e^{-2t} \begin{bmatrix} 2 \\ -3 \end{bmatrix}$
10. $\mathbf{x}' = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \mathbf{x} + e^t \begin{bmatrix} t \\ 1 \end{bmatrix}$
11. $\mathbf{x}' = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 5 \\ 0 & 5 & 1 \end{bmatrix} \mathbf{x} + e^t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
12. $\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix} \mathbf{x} + t \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$

In Problems 13–19, determine whether the given vector functions are linearly dependent (LD) or linearly independent (LI) on the interval $(-\infty, \infty)$.

13. $\begin{bmatrix} t \\ 3 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$
14. $\begin{bmatrix} te^{-t} \\ e^{-t} \end{bmatrix}$, $\begin{bmatrix} e^{-t} \\ e^{-t} \end{bmatrix}$
15. $e^t \begin{bmatrix} 1 \\ 5 \end{bmatrix}$, $e^t \begin{bmatrix} -3 \\ -15 \end{bmatrix}$
16. $\begin{bmatrix} \sin t \\ \cos t \end{bmatrix}$, $\begin{bmatrix} \sin 2t \\ \cos 2t \end{bmatrix}$
17. $e^{2t} \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$, $e^{2t} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$, $e^{3t} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
18. $\begin{bmatrix} \sin t \\ \cos t \end{bmatrix}$, $\begin{bmatrix} \sin t \\ \sin t \end{bmatrix}$, $\begin{bmatrix} \cos t \\ \cos t \end{bmatrix}$

$$19. \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} t \\ 0 \\ t \end{bmatrix}, \begin{bmatrix} t^2 \\ 0 \\ t^2 \end{bmatrix}$$

20. Let

$$\mathbf{x}_1 = \begin{bmatrix} \cos t \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} \sin t \\ \cos t \\ \cos t \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} \cos t \\ \sin t \\ \cos t \end{bmatrix}.$$

- (a) Compute the Wronskian.
- (b) Are these vector functions linearly independent on $(-\infty, \infty)$?
- (c) Is there a first-order homogeneous linear system for which these functions are solutions?

In Problems 21–24, the given vector functions are solutions to a system $\mathbf{x}'(t) = \mathbf{Ax}(t)$. Determine whether they form a fundamental solution set. If they do, find a fundamental matrix for the system and give a general solution.

21. $\mathbf{x}_1 = e^{2t} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $\mathbf{x}_2 = e^{2t} \begin{bmatrix} -2 \\ 4 \end{bmatrix}$
22. $\mathbf{x}_1 = e^{-t} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $\mathbf{x}_2 = e^{4t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
23. $\mathbf{x}_1 = \begin{bmatrix} e^{-t} \\ 2e^{-t} \\ e^{-t} \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} e^{3t} \\ -e^{3t} \\ 2e^{3t} \end{bmatrix}$
24. $\mathbf{x}_1 = \begin{bmatrix} e^t \\ e^t \\ e^t \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} \sin t \\ \cos t \\ -\sin t \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} -\cos t \\ \sin t \\ \cos t \end{bmatrix}$

25. Verify that the vector functions

$$\mathbf{x}_1 = \begin{bmatrix} e^t \\ e^t \end{bmatrix} \quad \text{and} \quad \mathbf{x}_2 = \begin{bmatrix} e^{-t} \\ 3e^{-t} \end{bmatrix}$$

are solutions to the homogeneous system

$$\mathbf{x}' = \mathbf{Ax} = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \mathbf{x},$$

on $(-\infty, \infty)$, and that

$$\mathbf{x}_p = \frac{3}{2} \begin{bmatrix} te^t \\ te^t \end{bmatrix} - \frac{1}{4} \begin{bmatrix} e^t \\ 3e^t \end{bmatrix} + \begin{bmatrix} t \\ 2t \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

is a particular solution to the nonhomogeneous system $\mathbf{x}' = \mathbf{Ax} + \mathbf{f}(t)$, where $\mathbf{f}(t) = \text{col}(e^t, t)$. Find a general solution to $\mathbf{x}' = \mathbf{Ax} + \mathbf{f}(t)$.

26. Verify that the vector functions

$$\mathbf{x}_1 = \begin{bmatrix} e^{3t} \\ 0 \\ e^{3t} \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} -e^{3t} \\ e^{3t} \\ 0 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} -e^{-3t} \\ -e^{-3t} \\ e^{-3t} \end{bmatrix}$$

are solutions to the homogeneous system

$$\mathbf{x}' = \mathbf{A}\mathbf{x} = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \mathbf{x}$$

on $(-\infty, \infty)$, and that

$$\mathbf{x}_p = \begin{bmatrix} 5t+1 \\ 2t \\ 4t+2 \end{bmatrix}$$

is a particular solution to $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f}(t)$, where $\mathbf{f}(t) = \text{col}(-9t, 0, -18t)$. Find a general solution to $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f}(t)$.

27. Prove that the operator defined by $L[\mathbf{x}] := \mathbf{x}' - \mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is an $n \times n$ matrix function and $\mathbf{x}$ is an $n \times 1$ differentiable vector function, is a linear operator.
28. Let $\mathbf{X}(t)$ be a fundamental matrix for the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$. Show that $\mathbf{x}(t) = \mathbf{X}(t)\mathbf{X}^{-1}(t_0)\mathbf{x}_0$ is the solution to the initial value problem $\mathbf{x}' = \mathbf{A}\mathbf{x}$, $\mathbf{x}(t_0) = \mathbf{x}_0$.

In Problems 29–30, verify that $\mathbf{X}(t)$ is a fundamental matrix for the given system and compute $\mathbf{X}^{-1}(t)$. Use the result of Problem 28 to find the solution to the given initial value problem.

29. $\mathbf{x}' = \begin{bmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$;

$$\mathbf{X}(t) = \begin{bmatrix} 6e^{-t} & -3e^{-2t} & 2e^{3t} \\ -e^{-t} & e^{-2t} & e^{3t} \\ -5e^{-t} & e^{-2t} & e^{3t} \end{bmatrix}$$

30. $\mathbf{x}' = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$;

$$\mathbf{X}(t) = \begin{bmatrix} e^{-t} & e^{5t} \\ -e^{-t} & e^{5t} \end{bmatrix}$$

31. Show that

$$\begin{vmatrix} t^2 & t|t| \\ 2t & 2|t| \end{vmatrix} \equiv 0$$

on $(-\infty, \infty)$, but that the two vector functions

$$\begin{bmatrix} t^2 \\ 2t \end{bmatrix}, \quad \begin{bmatrix} t|t| \\ 2|t| \end{bmatrix}$$

are linearly independent on $(-\infty, \infty)$.

32. **Abel's Formula.** If $\mathbf{x}_1, \dots, \mathbf{x}_n$ are any n solutions to the $n \times n$ system $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$, then Abel's formula gives a representation for the Wronskian $W(t) := W[\mathbf{x}_1, \dots, \mathbf{x}_n](t)$. Namely,

$$W(t) = W(t_0) \exp\left(\int_{t_0}^t \{a_{11}(s) + \dots + a_{nn}(s)\} ds\right),$$

where $a_{11}(s), \dots, a_{nn}(s)$ are the main diagonal elements of $\mathbf{A}(s)$. Prove this formula in the special case when $n = 3$. [Hint: Follow the outline in Problem 30 of Exercises 6.1, page 327.]

33. Using Abel's formula (Problem 32), confirm that the Wronskian of n solutions to $\mathbf{x}' = \mathbf{A}\mathbf{x}$ on the interval I is either identically zero on I or never zero on I .
34. Prove that a fundamental solution set for the homogeneous system $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$ always exists on an interval I , provided $\mathbf{A}(t)$ is continuous on I . [Hint: Use the existence and uniqueness theorem (Theorem 2) and make judicious choices for $\mathbf{x}_0$.]
35. Prove Theorem 3 on the representation of solutions of the homogeneous system.
36. Prove Theorem 4 on the representation of solutions of the nonhomogeneous system.
37. To illustrate the connection between a higher-order equation and the equivalent first-order system, consider the equation

$$(11) \quad y'''(t) - 6y''(t) + 11y'(t) - 6y(t) = 0.$$

- (a) Show that $\{e^t, e^{2t}, e^{3t}\}$ is a fundamental solution set for (11).
- (b) Using the definition in Section 6.1, compute the Wronskian of $\{e^t, e^{2t}, e^{3t}\}$.
- (c) Setting $x_1 = y, x_2 = y', x_3 = y''$, show that equation (11) is equivalent to the first-order system

$$(12) \quad \mathbf{x}' = \mathbf{A}\mathbf{x},$$

where

$$\mathbf{A} := \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}.$$

- (d) The substitution used in part (c) suggests that

$$S := \left\{ \begin{bmatrix} e^t \\ e^t \\ e^t \end{bmatrix}, \begin{bmatrix} e^{2t} \\ 2e^{2t} \\ 4e^{2t} \end{bmatrix}, \begin{bmatrix} e^{3t} \\ 3e^{3t} \\ 9e^{3t} \end{bmatrix} \right\}$$

is a fundamental solution set for system (12). Verify that this is the case.

- (e) Compute the Wronskian of S . How does it compare with the Wronskian computed in part (b)?
38. Define $\mathbf{x}_1(t)$, $\mathbf{x}_2(t)$, and $\mathbf{x}_3(t)$, for $-\infty < t < \infty$, by

$$\mathbf{x}_1(t) = \begin{bmatrix} \sin t \\ \sin t \\ 0 \end{bmatrix}, \quad \mathbf{x}_2(t) = \begin{bmatrix} \sin t \\ 0 \\ \sin t \end{bmatrix}, \quad \mathbf{x}_3(t) = \begin{bmatrix} 0 \\ \sin t \\ \sin t \end{bmatrix}.$$

- (a) Show that for the three scalar functions in each individual row there are nontrivial linear combinations that sum to zero for all t .
- (b) Show that, nonetheless, the three vector functions are linearly independent. (No single nontrivial combination works for each row, for all t .)
- (c) Calculate the Wronskian $W[\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3](t)$.
- (d) Is there a linear third-order homogeneous differential equation system having $\mathbf{x}_1(t)$, $\mathbf{x}_2(t)$, and $\mathbf{x}_3(t)$ as solutions?